

JIEUN SUNG

Ph.D. Candidate in Computational Biology and AI for Drug Discovery

Seoul, Republic of Korea | jieunsung928@gmail.com | [GitHub](#)

EDUCATION

Integrated M.S.-Ph.D. Program in Life Science, with a specialization in Bioinformatics

Expected completion: February 2027

Ewha Womans University

Department of Life Science, Systems Pharmacology Laboratory

Conducting computational research in single-cell transcriptomics, perturbation-response modeling, and AI-driven drug discovery.

B.S. in Chemistry, with a Double Major in Bioinformatics

Completion: August 2021

Ewha Womans University

Chemistry and Bioinformatics

Recipient of an Academic Excellence Scholarship.

RESEARCH INTERESTS

AI-driven drug discovery; single-cell perturbation modeling and transcriptomic response prediction; drug-target interaction prediction; virtual cell modeling; cellular reprogramming and gene regulatory networks; biological foundation models; graph neural networks.

PUBLICATIONS

1. **Sung, J.** and Kim, W. (2026). scDEBART: Predicting *in silico* Single-Cell Perturbation Responses via Large-Scale Differential Expression Learning. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*. [OpenReview](#).
2. **Sung, J.**, Kim, S., and Kim, W. TDEP: Predicting Target Perturbation Response from Drug-Induced Transcriptomes. Manuscript under review.
3. **Sung, J.***, Son, K.*, Cha, H.-J., Kim, W., and Lee, H. FateDriver: A Dual-View Graph Neural Network for Predicting Driver Genes in Transcription Factor-Induced Cellular Reprogramming. Manuscript in preparation.
4. **Sung, J.** and Kim, W. TDEM: Deconvolving Drug-Induced Transcriptomes into Target-Specific Perturbation Profiles via Matrix Factorization. Manuscript in preparation.

* Equal contribution.

PATENTS

1. **Sung, J.** and Kim, W. Large-Scale Pretrained Transformer Model and Method for Predicting Expression Changes (logFC) from Single-Cell Transcriptomes. Republic of Korea provisional patent application No. 10-2026-0094556, filed May 26, 2026. Assignee: Ewha Womans University. Original title: 단일세포 전사체 기반 발현 변화(logFC) 예측을 위한 대규모 사전학습 트랜스포머 모델 및 그 방법.

HONORS & AWARDS

Jul-Aug 2025	Third Place, 3rd AI Drug Discovery Competition Placed 3rd among 1,137 participants in a national competition focused on predicting MAP3K5 inhibitor IC50 values from molecular structures. Hosted by the Korea Pharmaceutical and Bio-Pharma Association and sponsored by the Ministry of Health and Welfare, Yuhan Corporation, and CAS.
Aug-Sep 2024	Second Place, 2nd AI Drug Discovery Competition Placed 2nd among 1,798 participants in a national competition focused on predicting IRAK4 inhibitor IC50 values from molecular structures. Hosted by the Korea Pharmaceutical and Bio-Pharma Association's AI Drug Convergence Research Institute and sponsored by the Ministry of Health and Welfare, Korea Health Industry Development Institute, and Daewoong Pharmaceutical.

RESEARCH FUNDING

National Research Foundation of Korea (NRF) Ph.D. Student Research Promotion Grant
AI-Based Modeling of Drug-Induced Transcriptomic Changes
Funding period: Sep 2024–Aug 2026
Principal Investigator | Total funding: KRW 50,000,000.

SELECTED RESEARCH PROJECTS

scDEBART | *ICML 2026*

Built a large-scale Transformer foundation model (BART encoder–decoder) for single-cell perturbation-response prediction, reframing the pretraining objective from reconstructing absolute expression to predicting log fold-changes conditioned on basal expression. Curated a pretraining corpus of 6.28 million differential expression profiles from 66.6 million cells across 617 datasets, and pretrained the model on $4 \times$ NVIDIA H100 GPUs using DeepSpeed ZeRO-3 (452 GPU-hours). The model achieved 4–7 $\times$ higher recovery of top-ranked perturbation-responsive genes than leading baselines across diverse CRISPRi and CRISPRa datasets, and 71.4% top-1 accuracy on reverse perturbation identification versus near-zero for prior models.

TDEP | *Under Review*

Developed a framework that maps drug-induced transcriptomic signatures to target-specific perturbation responses, enabling *in silico* inference of gene-level effects from large-scale pharmacological perturbation data.

FateDriver | *Manuscript in Preparation*

Designed a dual-view graph neural network that jointly models cell-state transitions and gene regulatory structure to identify candidate driver genes underlying transcription factor-induced cellular reprogramming across multiple single-cell datasets.

Sequence-Similarity–Guided Multi-Representation Ensemble for MAP3K5 IC50 Prediction | *AI Drug Discovery Competition*

Developed an uncertainty-aware ensemble of graph-, SMILES-, and SELFIES-based molecular representations. Combined predictive uncertainty with PSI-BLAST-derived protein sequence similarity to adaptively weight ensemble members.

PRESENTATIONS

Invited Talks

1. **Sung, J.** Building AI Prediction Models from Biological Data. Invited lecture, Systems Biology Lab, Department of Pharmacy, Pusan National University, Busan, Republic of Korea, February 2026. Two-and-a-half-hour tutorial with open discussion for graduate researchers.

Contributed Talks and Posters

1. **Sung, J.**, Son, K., Cha, H.-J., Kim, W., and Lee, H. FateDriver: A Dual-View Graph Neural Network for Predicting Driver Genes in TF-Induced Cellular Reprogramming. Annual Meeting of the Korean Society for Stem Cell Research (KSSCR), Republic of Korea, August 2026. Poster.
2. **Sung, J.** and Kim, W. scDEBART: Predicting *in silico* Single-Cell Perturbation Responses via Large-Scale Differential Expression Learning. 43rd International Conference on Machine Learning (ICML 2026), COEX Convention & Exhibition Center, Seoul, Republic of Korea, July 2026. Poster.
3. **Sung, J.**, Son, K., Cha, H.-J., Kim, W., and Lee, H. FateDriver: A Dual-View Graph Neural Network for Predicting Driver Genes in TF-Induced Cellular Reprogramming. Annual Meeting of the Korean Society for Aging Research, Republic of Korea, June 2026. Poster.
4. **Sung, J.**, Kim, S., and Kim, W. TDEP: Target-Perturbed Differential Expression Profiles for Interpretable Drug–Target Interaction Prediction. Annual Meeting of the Korean Society for Bioinformatics (KSBI), Republic of Korea, October 2025. Poster.

5. **Sung, J.** and Kim, W. Uncertainty-Aware BLAST-Weighted Multi-Representation Ensemble Model for IC50 Prediction. 3rd AI Drug Discovery Competition, Republic of Korea, September 2025. Poster.
6. **Sung, J.**, Kim, S., and Kim, W. AI-Based Deconvolution of Drug-Induced Transcriptomic Changes into Single-Target Effects across Multiple Cell Lines. EJO Symposium, Republic of Korea, December 2024. Poster and oral presentation.
7. **Sung, J.** and Kim, W. Ensemble Learning-Based Prediction of IRAK4 Inhibitory Activity (IC50). 2nd AI Drug Discovery Competition, Republic of Korea, October 2024. Poster.
8. **Sung, J.** and Kim, W. Deconvolution of Transcriptomic Changes Caused by Drug Perturbation Using Matrix Factorization. Annual Meeting of the Korea Genome Organization (KOGO), Republic of Korea, January–February 2024. Poster.
9. **Sung, J.** and Kim, W. Deconvolution of Transcriptomic Changes Caused by Drug Perturbation Using Matrix Factorization. Annual Meeting of the Korean Society for Bioinformatics (KSBI), Republic of Korea, November 2023. Poster.
10. **Sung, J.** and Kim, W. Deconvolution of Transcriptomic Changes Caused by Drug Multi-Target Perturbation Using Matrix Factorization. Annual Meeting of the Korean Society for Biochemistry and Molecular Biology (KSBMB), Republic of Korea, May 2023. Poster.
11. **Sung, J.** and Kim, W. Deconvolution of Transcriptomic Changes Caused by Drug Multi-Target Perturbation Using a Neural Network. Annual Meeting of the Korean Society for Bioinformatics (KSBI), Republic of Korea, October 2022. Poster.

TECHNICAL EXPERTISE

Programming: Python, R, Bash; Git, LaTeX

Machine Learning & AI: PyTorch, Hugging Face Transformers, DeepSpeed; Transformer architectures, graph neural networks, deep generative models, self-supervised pretraining, multi-GPU distributed training, biological foundation models

Statistics & Probabilistic Modeling: Variational inference, Bayesian differential expression, uncertainty quantification, uncertainty-weighted ensembling

Computational Biology: Single-cell RNA-seq analysis (scanpy, anndata, scvi-tools), perturbation transcriptomics (Perturb-seq, CRISPRi/CRISPRa), differential expression analysis, gene regulatory network inference, pathway enrichment

Computational Drug Discovery: Molecular property prediction, molecular representation learning (graph, SMILES, SELFIES), drug–target interaction prediction, molecular docking, virtual screening